Jaron Droder

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EDUCATION

University of Colorado Boulder — May of 2026

B.A. Physics | B.A. Mathematics, Computational Focus

Minor: Atmospheric and Oceanic Sciences

RESEARCH AND PROFESSIONAL EXPERIENCE

Undergraduate Researcher, REU in Topological Data Analysis at CU Boulder, Summer 2025

Mentor: Prof. Markus Pflaum

- Built tools to process structured mathematical data and extract torsion invariants and Betti numbers with emphasis on reproducibility more efficient algorithms.
- Applied computational methods to molecular conformational landscapes and potential energy surfaces, rapidly building familiarity with unfamiliar mathematical and physical models.
- Used SSH workflows, technical documentation, and Git to organize code practices to manage collaboration and communicate research methods clearly.

Quality Assurance Tester at PhET Interactive Simulations, CU Boulder. November 2022 to September 2024

- Produced detailed technical reports documenting software issues, reproduction steps, system behavior, and expected versus observed outcomes for software engineering teams.
- Tested and debugged interactive scientific simulations across varied software environments, developing strong system level troubleshooting and analytical skills.

Production Equipment Manager at Program Council of CU Boulder, November 2024 to May 2026

- Led event production for large live events. This included technical setup and troubleshooting while handling lighting, audio, staging, and specialized equipment.
- Communicated technical workflows clearly with performers, crew members, and clients in time sensitive environments.

PROJECTS

Hypothetical Cloud Seeding Experiment Design — Mountain Meteorology, CU Boulder, Spring 2026.

- Designed a hypothetical paired basin research study to evaluate potential winter cloud seeding impacts on snowpack in Idaho's Snake River Basin.
- Compared target and control watersheds using orographic precipitation patterns, storm flow direction, basin topography, and snowpack measurement strategy.

Analog Signal Processing Circuit / Octave Pedal — Electronics for Physics Laboratory, Spring 2026

- Designed and tested a multi stage analog signal processing circuit using op-amp buffers, rectifiers, filters, comparators, AC coupling, and potentiometer voltage dividers.
- Tuned gain, filtering, and voltage divider stages to match signal amplitudes and improve output stability

TECHNICAL SKILLS

- **Languages & Tools:** C++, Python, Julia, Git, Mathematica, LaTeX, Maple
- **Competencies:** Algorithm design, numerical methods, computational topology, data analysis, debugging complex systems, technical writing
- **Technical Areas:** Electrical Circuits, Scientific Computing, Data Analysis, Modeling, Numerical Methods

RELEVANT COURSEWORK

- **Computer Science:** Data Structures, Algorithms
- **Physics:** Experimental Physics I and II, Modern Physics, Classical Mechanics I–II, Electricity & Magnetism I–II, Quantum Mechanics I, Atmospheric Modeling, Physical Oceanography, Atmospheric Physics, Mountain Meteorology, Electronics for Physics, Thermodynamics and statistical mechanics.
- **Mathematics:** Calculus I–III, Differential geometry, Linear Algebra, Differential Equations, Discrete Math, Real Analysis, Combinatorics, Topology, Intermediate Numerical Analysis